🚀 Hysteresis 26–45 Python Projects

Python 3.8+ | Projects 19 | Algorithms Data Structures | Systems Distributed | License Open Source

A curated collection of **Python system design and algorithm projects** ranging from:

- Data Structures
- Search Engines
- Distributed Systems
- Backend Simulation
- Game Engines
- Kubernetes Scheduler Simulation

All projects are **standalone**, lightweight, and require **no external dependencies**.

📚 Project Catalog

ID	Project	Category
27	Autocomplete (Trie)	Data Structures
28	File System Simulator	System Design
29	Maze Generator & Solver	Algorithms
30	Mini Search Engine (TF-IDF)	Information Retrieval
31	Social Graph	Graph Algorithms
32	Mini Git	Version Control
33	Log Analyzer	Data Processing
34	Spell Checker	NLP
35	Sudoku Solver	Backtracking
36	Expression Evaluator	Parsing
37	Task Scheduler	OS Concepts
38	Memory Allocator	Operating Systems
39	URL Shortener	Backend Design
40	Stock Analyzer	Data Analysis
41	Distributed Crawler & Search	Distributed Systems
42	Ride Sharing Backend	System Design
43	Distributed KV Store	Databases

ID	Project	Category
44	Multiplayer Game Engine	Networking
45	Kubernetes-like Scheduler	Cloud Systems

🗂 Project Folder Structure

```
python-projects/
│
├── 27_autocomplete.py
├── 28_filesystem.py
├── 29_maze.py
├── 30_search_engine.py
├── 31_social_graph.py
├── 32_mini_git.py
├── 33_log_analyzer.py
├── 34_spell_checker.py
├── 35_sudoku_solver.py
├── 36_expression_eval.py
├── 37_task_scheduler.py
├── 38_memory_allocator.py
├── 39_url_shortener.py
├── 40_stock_analyzer.py
│
├── 41_crawler/
│   ├── main.py
│   ├── crawler.py
│   ├── index.py
│   └── seeds.py
│
├── 42_rideshare/
│   ├── main.py
│   ├── models.py
│   └── service.py
│
├── 43_kv_store/
│   ├── server.py
│   ├── client.py
│   └── storage.py
│
├── 44_game_engine/
│   ├── main.py
│   ├── rooms.py
│   └── player.py
│
└── 45_scheduler/
    ├── main.py
    ├── node.py
    └── pod.py
```

⚙️ How to Run

Single-file projects

```
python 27_autocomplete.py
python 28_filesystem.py
python 35_sudoku_solver.py
```

Folder-based projects

```
python -m 41_crawler.main
python -m 42_rideshare.main
python -m 44_game_engine.main
```

🧠 System Architecture Overview

Distributed Crawler & Search Engine (Project 41)

```
          +-------------+
          |  Seed URLs  |
          +------+------+
                 |
                 v
           +-----+------+
           |  Crawler   |
           +-----+------+
                 |
                 v
           +-----+------+
           | HTML Parser|
           +-----+------+
                 |
                 v
           +-----+------+
           |  Indexer   |
           +-----+------+
                 |
                 v
           +-----+------+
           | Search API |
           +------------+
```

Workflow

1. Seed URLs start the crawl
2. Pages are fetched
3. Links are extracted
4. Text content is indexed
5. Queries return ranked results

Distributed Key-Value Store (Project 43)

```
          +-----------+
Client -->|  Server   |
          +-----+-----+
                |
                v
          +-----+-----+
          | In-Memory |
          |  Storage  |
          +-----------+
```

Supports operations:

- SET key value
- GET key
- DELETE key
- TTL expiration
- Pub/Sub messaging

Kubernetes-like Scheduler (Project 45)

```
           +----------+
           |  Pods    |
           +----+-----+
                |
                v
          +-----+-----+
          | Scheduler |
          +-----+-----+
                |
      +---------+----------+
      |         |          |
   Node-1    Node-2     Node-3
```

Scheduler assigns pods to nodes based on:

- CPU capacity
- Memory availability
- Node load balancing

📊 Example Outputs

Autocomplete

```
['python', 'pytorch', 'pycharm', 'pyramid', 'py']
```

Maze Solver

```
Path length: 85
```

URL Shortener

```
KrthEZ -> https://example.com
```

Distributed Crawler

```
Mini search ready. Type a query (or 'exit').

> link.springer.com
1. https://link.springer.com (score=6.000)
2. https://www.springer.com (score=4.000)
```

Distributed Crawler & Search (Project 41)

Run:

python -m 41_crawler.main

Output:

Mini search ready. Type a query (or 'exit').

> link.springer.com

1. https://link.springer.com (score=6.000)
2. https://www.springer.com (score=4.000)
3. https://gemini.google.com (score=2.000)
4. https://www.anthropic.com (score=2.000)
5. https://platform.openai.com (score=2.000)

> mits.ac.in

1. https://nptel.ac.in (score=4.000) 🚖 Ride Sharing Backend (Project 42) Ride requested Driver assigned Ride started Ride completed Fare: 20.93 🎮 Multiplayer Game Engine (Project 44) Rooms: ['room-1', 'room-2'] 🎡 Kubernetes-like Scheduler Simulator (Project 45) Node-1 -> Pod A Node-2 -> Pod B Node-3 -> Pod C 🎯 Learning Goals

These projects demonstrate:

Data Structures

Algorithms

Backend Architecture

Distributed Systems

Search Engine Design

Graph Algorithms

Cloud Scheduling Concepts

⚙️ Requirements

Python 3.8+

Check version:

python --version ⭐ Purpose

This repository acts as a Python backend & system design portfolio containing 19 practical projects covering core software engineering concepts.